

Date: _____

Week 9

October 19, 2026 • 5:00 – 7:30 PM • Kai Santos

NOTES

MECH 191 — Metallurgy · Week 9 Notes · fill in blanks & key terms as you go

- 01 Standards, Codes & Defect Recognition**
ASTM, AWS, ASME, ISO, and API — what each governs; manufacturing and service defects and how each is identified
- 02 Indication → Defect → Reject**
Precise definitions and the evaluation sequence from detection to final disposition
- 03 Inspection Documentation**
What a valid inspection report must contain, traceability, record retention, and non-conformance reports

NOTES

- Alloy A — 1018 steel (normalized)

Portfolio Handout — Module 8

NOTES

MECH 191 — Metallurgy · Week 9 Notes · fill in blanks & key terms as you go

MECH 191 — Course & Unit Roadmap

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Materials Fundamentals	Steel & Alloys	Testing & Quality	Manufacturing	Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 ← here Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

SLIDE 4 MECH 191 — Course & Unit Roadmap

NOTES

Topics

- ## Resources

Moniz Ch. 10
Materials Standards

- Quality Standards & Codes
- Reading Inspection Documentation
- Defect Recognition

Covers Weeks 7–9
Scheduled this session

SLIDE 5

NOTES

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- ## SLIDE 6 Core Questions

Quality Standards & Codes — Knowing Which Rules Apply

Major Standards Bodies			
Body	Governs	Key Standard(s)	Technician Relevance
ASTM	Material specs & test methods	A36, A370, E165	Acceptance criteria for base metals; test procedures for mechanical and NDT
AWS	Welding codes & inspection	D1.1 Structural Steel	Weld acceptance criteria, inspection requirements, welder qualification
ASME	Pressure vessels, boilers & piping	BPVC Sec. V (NDT), Sec. IX	NDT methods and procedures; welding qualification for pressure equipment
ISO	International standards (global use)	9606, 17636, 17640	Welder qualification, RT and UT of welds — used on international projects
API	Oil & gas industry	1104, 570, 650	Pipeline welding, piping inspection, storage tank fabrication and repair

Using the wrong standard is as serious an error as missing a defect.

Acceptance criteria differ between codes — a weld accepted under AWS D1.1 may be rejectable under ASME Section VIII, or vice versa. Know your governing document before you inspect.

SLIDE 7 *Quality Standards & Codes — Knowing Which Rules Apply*

NOTES

Porosity

Slag Inclusions

Lack of Fusion (LOF)

Cause: Insufficient heat input, incorrect travel speed, or poor technique — weld metal does not fuse with base metal or previous pass.

Appears: Linear indication parallel to fusion face on RT (may be missed if beam isn't parallel); strong planar reflector on UT. Not detectable by PT/MT unless surface-breaking.

Risk: Pre-existing planar crack with a sharp tip — most standards have zero tolerance for LOF.

Lack of Penetration (LOP)

Cause: Insufficient heat or incorrect joint preparation — incomplete fusion at the weld root.
Appears: Continuous or intermittent linear indication at weld root centerline on RT; detectable by UT; visible to VT on root side if accessible.
Risk: Reduces effective throat and creates a sharp root notch — serious in cyclically loaded structures.

SLIDE 8 Common Weld Defects — Manufacturing Discontinuities

NOTES

Fabrication Defects (cont.)

Hot cracks (solidification), cold/hydrogen-induced cracks (HAZ, <300°C), lamellar tearing, crater cracks. Appear as fine linear indications on RT, linear bleed-out on PT/MT, strong planar reflectors on UT. A crack is rejectable in virtually all standards regardless of size — no minimum applies.

Groove at the weld toe caused by excessive current or incorrect technique. Visible to VT; detectable by PT. Creates a stress concentration especially dangerous in fatigue-loaded structures. Acceptance criteria limit depth and length.

Weld metal rolled over base metal without fusing — low current or slow travel. Visible to VT as a lip at the toe; detectable by PT. Creates a sharp notch similar to undercut. No acceptance criteria — no overlap is acceptable.

Initiate at stress concentrations (notches, weld toes, holes, surface scratches) and propagate perpendicular to the applied cyclic stress. Often invisible to VT until large — PT, MT, or UT required for early detection.

Branching crack networks forming in susceptible alloys under the simultaneous presence of stress and a corrosive environment. Common in austenitic stainless (chlorides) and high-strength steels. Appears as branching linear PT indications.

Occurs at elevated temperature under sustained stress — grain boundary cavitation progresses to cracking over time. Visible metallographically in cross-section; later-stage damage may be detectable by UT or RT.

Localised material loss — pits act as stress concentrations and can initiate fatigue or SCC cracks. VT reveals visible pits; UT thickness surveys map remaining wall. Severity assessed by pit depth and density.

NOTES

MECH 191 — Metallurgy · Week 9 Notes · fill in blanks & key terms as you go

Indication → Discontinuity → Defect → Reject — Precise Definitions

1 Indication	2 Discontinuity	3 Defect	4 Reject / Disposition
Definition: Any response or evidence of a discontinuity detected by an NDT method. Simply means something was found — no judgment about significance. Key point: Non-relevant indications (caused by geometry — weld caps, threads, edges) must be distinguished from real discontinuities.	Definition: A physical interruption in the material — a void, crack, inclusion, or boundary — that may or may not be harmful. Key point: Every defect is a discontinuity, but not every discontinuity is a defect. Evaluation against criteria is required.	Definition: A discontinuity that has been evaluated against the acceptance criteria of the applicable standard and found to exceed the allowable limits. Key point: The word 'defect' has a legal meaning — it implies a finding of non-conformance. Do not use it casually before evaluation is complete.	Definition: The disposition applied to a component containing one or more defects — repair, rework, scrap, or use-as-is with engineering approval (fitness for service). Key point: Finding an indication does not equal rejection. Every indication must be evaluated against acceptance criteria before a disposition is made.

SLIDE 10 *Indication → Discontinuity → Defect → Reject — Precise Defin*

NOTES

Minimum Report Content

- Component ID: drawing number, heat/lot number, weld ID or location reference
- Applicable standard and revision
- NDT method and written procedure reference number
- Equipment used: make, model, serial number, calibration due date
- Inspector name, certification level, and certification scheme
- Date, time, and location of inspection
- Environmental conditions (temperature, lighting for VT, surface condition)
- Description of any indications: location, orientation, size/length, detection method
- Acceptance criteria applied and evaluation result for each indication
- Final disposition: accept or reject, with repair or re-inspection requirements
- Inspector signature and date

Every record must link to the physical component through unique identifiers — without traceability the record is useless in an audit or failure investigation.

In most regulated industries records must be retained for the design life of the equipment. Electronic records must be backed up and protected against alteration.

When a component is rejected, a formal NCR is raised documenting:

- Nature of the non-conformance
- Disposition: repair, rework, scrap, or use-as-is with engineering approval
- Re-inspection requirements after repair

Verbal assurances are worthless in a quality audit — if it isn't documented, it didn't happen.

NOTES

Technician's Perspective — Documentation and Disposition in Practice

Red Flag: Report missing procedure reference and inspector certification

An inspection report arrives describing a linear PT indication within acceptance limits — but no procedure number, no equipment calibration record, and no inspector certification level are listed. Do not accept the component. Return the report, require complete documentation, and arrange re-inspection if procedural compliance cannot be demonstrated. The finding is meaningless if the inspection cannot be shown to have been performed correctly.

Caution: Indication evaluated against the wrong standard

A weld in a pressure vessel is evaluated against AWS D1.1 (structural steel) acceptance criteria. The applicable code is actually ASME Section VIII. Acceptance criteria differ between codes — a finding that passes AWS D1.1 may exceed ASME limits. This is as serious as missing a defect: confirm the governing standard before evaluating any indication.

Good Practice: Complete NCR triggers correct disposition

A root crack is found in a pipeline girth weld by UT. A Non-Conformance Report is raised immediately, the weld is marked and quarantined, and re-inspection after repair is scheduled with the same qualified procedure and inspector level. The NCR provides a traceable record of the finding, the repair, and the re-test — essential documentation if the line ever fails in service.

SLIDE 12 *Technician's Perspective — Documentation and Disposition in*

NOTES

What is a quality standard and why does it exist?

What are the most common defects a technician will encounter?

Indication vs. discontinuity vs. defect vs. reject?

What does good inspection documentation look like — and why does an incomplete report matter?

SLIDE 13 *Core Questions — Answers*

NOTES

1

2

3

SLIDE 14 Key Takeaways

NOTES

Week 9 Resources

Materials Standards — primary reading

ASTM, AWS, ASME, ISO, API reference summary

Sample MTR, inspection report walkthrough

Visual identification of all common weld defects

Covers Weeks 7-9 — Mechanical Testing, NDT, Standards

October 26, 2026 · Unit 4: Manufacturing begins

- Sand casting, investment casting, and die casting — how each process works and what it produces
- How solidification microstructure differs from wrought microstructure — and why it matters for properties
- Common casting defects — porosity, shrinkage, hot tears — and how to identify them
- Why castings require Brinell hardness testing and often need heat treatment before machining

NOTES